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RESEARCH PROPOSAL

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Project title: ***Context-Aware NLP Models for Detecting and Mitigating Sensitive Information in Higher Education AI Prompts***

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1. Proposed Research Title

Context-Aware NLP Models for Detecting and Mitigating Sensitive Information in Higher Education AI Prompts

2. Statement of the Research Problem

The adoption of AI tools such as Chat-GPT in higher education has become a fixture in university life, and students use them constantly for assignments, research and even for personal reflection. What most students do not realize is that every prompt they type may contain personal details, academic data or institutional information that gets transmitted to an external server the moment they hit send. There is currently no system that checks what a student is about to share before it leaves their device. While studies such as Arp et al. (2024) and Miresghallah et al. (2024) have proposed prompt sanitation solutions, these are designed for the enterprise environments and do not account for student users in higher education. If this gap remains unaddressed students will continue to unknowingly expose personal and institutional data, students and institutions we remain exposed to South Africa's Protection of Personal Information Act (POPIA) and an entire generation will enter the workforce without having ever developed safe AI interaction habits.

3. Background of the Research Problem

Artificial Intelligence tools have become increasingly embedded in everyday academic life, with students across the globe using platforms such as Chat-GPT, Google Gemini and Microsoft Copilot to assist with assignments, research and learning activities. This is not a bad thing. While these tools offer them significant benefits, they simultaneously introduce a largely unacknowledged risk. Students are among the most frequent and the least cautious users of AI (Yan et al. 2024). When students type a prompt, they often share a lot more than they intend to.

Unlike clicking a button or just filling in a form, AI interactions are open-ended. Students write naturally the way they think. That means that they drop their name, their student number, health condition, financial situation, sometimes all in one single prompt asking for help with essay. Kim et al. (2024) found that more than 70% of over one million real chatbots conversations contained personally identifiable information. That is not a small edge case. That is the norm.

A high-profile real-world incident at Samsung Electronics made this concrete. Employees submitted confidential source code and internal meetings noted to Chat-GPT. Resulting to an irreversible leakage of proprietary data to Open-AI's servers, Samsung responded to this by banning the use of AI tools entirely (Bloomberg 2023). That one option but banning the tools that students genuinely benefit from is not a real solution. South African higher education institutions face an additional layer of complexity because the Protection of Personal Information Act (POPIA) requires institution to protect personal information, yet there is no technical standard currently which governs how students interact with third-party AI tools compliance with this legislation (Roos, 2024).

Researchers have attempted to address related aspects of this problem. Mireshghallah et al (2024) proposed Casper, a prompt sanitation system that detects and redacts sensitive tokens before submission to a web-based Large Learning Model (LLM). Chen et al. (2025) went further with Preempt, using encryption and differential privacy to protect user data at inference time. Both are very invaluable technical contributions, but neither of them was built with a student in mind, and neither considers what it means to write a prompt in an academic context, where sensitives it not only about names and ID numbers but about the broader content of what is being shared and why? Sharma et al. (2024) and Contextual AI Lab (2025) demonstrate that context is often determined by the context rather than content alone. A sentence may look harmless in isolation but can be sensitive depending on what surrounds it, a dimension that current tools fail to adequately address.

The gap that remains therefore is clear. No existing system has been designed specifically to detect and mitigate sensitive information in student AI prompts within a higher education environment, using context-aware NLP that understands the academic and institutional setting of the user. This research aims to address that gap directly, proposing a system that not only detects sensitive content but also intervenes through warning. Prompts rephrasing and risk acknowledgment mechanisms that empower students to engage with AI tools more responsibly.

4. Research Aim, Objectives and Questions

❖ Research Aim

To develop a context-aware NLP system capable of detecting and mitigating sensitive information in student AI prompts within higher education institutions, in order to protect student data and promote responsible AI usage.

❖ Research Objectives

1. To assess the level awareness among higher education students regarding the risks of sharing sensitive information through AI prompts.
2. To investigate the types of sensitive information that students commonly include in AI prompts within the higher education environments
3. To implement mitigation mechanisms that protect students from unintentional data exposure.
4. To propose a context-aware NLP model capable of detecting sensitive information in student AI prompts in real time.

❖ Research Questions

1. How are higher education students aware of the privacy risks associated with sharing sensitive information when interacting with AI tools?
2. What types of sensitive information do students most frequently expose when interacting with AI tools in academic settings?
3. What mitigation strategies can be integrated into an NLP system to effectively prevent sensitive information from being submitted to AI tools by students?
4. How can a context-aware NLP model be designed to accurately detect sensitive information within student AI prompts?

5. Clarifications of terms

Term	General Definition	Study-Specific Context
Artificial Intelligence (AI)	The simulation of human intelligence processes by computer systems, including learning, reasoning, and self-correction (Russell and Norvig, 2021).	Refers specifically to large language model based tools such as ChatGPT, Google Gemini, and Microsoft Copilot that students interact with through natural language prompts in academic settings.
Natural Language Processing (NLP)	A branch of artificial intelligence concerned with enabling computers to understand, interpret, and generate human language in a meaningful way (Jurafsky and Martin, 2023).	Refers to the computational techniques applied to analyse and classify student prompt text in order to identify sensitive information before it is submitted to an AI tool.
Context-Aware System	A system that uses information about its environment and the circumstances of an interaction to adapt its behaviour accordingly (Dey, 2001).	Refers to the ability of the developed NLP system to evaluate not only the specific words in a student prompt but also the academic and institutional context in which those words appear, enabling more accurate sensitivity detection than keyword-based approaches alone.
Sensitive Information	Any data whose disclosure could cause harm, embarrassment, or legal consequences to an individual or organisation (Regulation (EU) 2016/679).	Includes personally identifiable information, academic records, health information, financial details, and institutional data that students may knowingly or unknowingly include in AI prompts.
Personally Identifiable Information (PII)	Any data that can be used to identify a specific individual, either on its own or in combination with other information (NIST, 2010).	Includes student names, identity numbers, student numbers, email addresses, contact details, and any information that could be used to identify a student submitting a prompt to an AI tool.
AI Prompts	A natural language input submitted by a user to an AI language model to elicit a response (Brown et al., 2020).	Text inputs submitted by higher education students during assignments, research, and studying.
Prompt Sanitization	Identifying and removing/modifying sensitive content before transmission to an AI provider (Mireshghallah et al., 2024).	Includes redaction, prompt rephrasing, and user warnings as alternative mitigation strategies.
Higher Education	Post-secondary education (universities/colleges) leading to degrees or diplomas (Council on Higher Education, 2016).	Specifically the South African environment, focusing on students at universities of technology like CPUT.
POPIA	South Africa's primary data protection legislation (Republic of South Africa, 2013).	Legal obligations for HEIs to protect student info shared inadvertently through AI tool interactions.
Mitigation	Reducing the severity, seriousness, or painfulness of something (Oxford English Dictionary, 2024).	Technical mechanisms (real-time warnings, rephrasing, risk acknowledgments) built into the proposed NLP system.

6. Literature Review

Students interact with AI tools every day and in doing so they share far more than they realize. The literature reviewed here presents discussions related to this issue from several angles, including ethical challenges posed by using artificial intelligence (AI) in education; the legal responsibilities of South African Colleges/Universities; technical systems/solutions that have been developed to mitigate prompt-based privacy issues; and the scientific methods used to detect the presence of context-sensitive information in regards to privacy. Collectively these sources demonstrate not only that the problem exists, but also the one particular gap that has been created because of this research has not been filled by anyone to date.

❖ The Ethical Risk Landscape in Higher Education

The work of Huang (2023) was one of the earliest, and clearest, accounts regarding what AI adoption in education means to student privacy. In the context of the ethics of AI being employed in education, Huang documents how utilizing an AI platform exposes student data through what she calls, secondary exploitation of an individual's personal information. So, once (the student) submits (any data) at an AI platform, this data can be altered, stored and then used for purposes that have nothing to do with the original task (the student had in mind). The risks of this practice are not theoretical; Huang points to data trafficking as a documented case of network fraud in which PII from an AI platform is trafficked for use by outside parties. In her paper, she recommends implementing stronger privacy protection systems; however, she fails to identify a technical solution for when the data leaves the student. This is her point of failure.

Extending the analysis of AI ethics in education, Mishara (2024) examines the broader AI in education accountability infrastructure and how those structures fail the students at the most fundamental level. According to Mishara, AI tools are designed to collect a notable amount of data of which the data collected exceeds the needs required of the task that it is performing, which in turn creates a surveillance infrastructure of educational technology. Currently, there is an accountability gap with no clear party responsible for the consequences of errors made by AI developers and educational institutions in their use of AI software. Students, as the most vulnerable group in this scenario, bear the greatest burden when something goes wrong. Mishara is calling for a new type of governance that would provide more oversight but fails to indicate what technical solutions might be available to address students' exposure to AI systems before any negative incidents occur.

Jiang (2024) takes a legal philosophy perspective and argues that ChatGPT poses challenges to an information subject's right to know throughout the entire user experience with the system: training data may have ambiguous sources, the manner in which user input is processed is not transparent, inferences drawn from the user data are not available to the user, and mechanisms for providing consent at the time of user interaction are essentially nonexistent. Jiang's proposal calls for countermeasures to be developed through cooperation between industries, regulatory agencies, and the courts. While Jiang is explicitly clear about what should happen, his analysis does little to provide a roadmap of how to implement it.

❖ The South African Legal Context

Davis and Trott (2024) offer the most directly relevant legal analysis for this research, published in the first edition of the African Journal on Privacy and Data Protection, which addresses the application of POPIA to AI systems in South Africa. Their conclusion is

straightforward: data protection laws are the only legally binding domestic law in the majority of African countries that govern any aspect of automated decision-making, and yet are insufficient to provide full protection from all privacy risks associated with AI systems. They identify three major gaps in POPIA's coverage with respect to the application of AI: (1) POPIA does not adequately address issues surrounding automated decision-making; (2) the repurposing of data for secondary purposes, which occurs at any time an AI model is fine-tuned using student data; and (3) the cross-border flow of data when a student in South Africa submits an instruction to a US-based AI platform, because this results in the student's data leaving South African jurisdiction and therefore being outside of the scope of POPIA.

For this study, the Davis and Trott analysis have clearly defined the institutional obligation to protect student personal information under POPIA at all South African universities. When students submit sensitive information through third-party AI tools there is no way for universities to fulfil their obligations, not because they do not care about meeting their obligation but rather there is no technical way to intervene before personal information leaves the university's possession and is sent to third parties. There exists a legal obligation but no technical means to satisfy that legal obligation.

❖ How Students Actually Behave with AI Tools

The Mireshghallah et al. (2024) twice-study represents the most empirical contribution to this body of literature through more than a million real world user conversation snippets collected from the WildChat dataset, exploring at what point and why people are willing to share personal identifiable information (PII) with AI chatbots. The study produced surprising results showing that people are willing to share sensitive PII not only without conscious intent but also with significantly higher frequency than anticipated given that most of the time users were completely focused doing something other than sharing their PII with AI chatbots. Sensitive PII was noted in 48% of translation assignment data and in 16% of code editing assignment data. Individuals did not intentionally choose to disclose their PII, they used natural language writing similar to the way in which people write when they need assistance completing tasks. PII disclosure was the result of memory retrieval failure in addition to producing more PII than they intended to when using natural language.

Mireshghallah et al. (2024) created a taxonomy based on qualitative and quantitative analyses of the conversations to fully capture the scope of disclosures. The taxonomy captures a much wider range of sensitive topics than just standard PII. Examples include detailed sexual preferences, certain drug use behaviors, immigration status, mental health disclosures, and financial situations types of information users disclosed during interactions related to tasks that had no connection to those topics. For example, if a student

is asking for editing help with their visa application, they could/may disclose their immigration history. When asking for help completing a personal statement, they could/may disclose their mental health status. A standard PII detector (which identifies PII like a name, phone number, or email address) would miss the entire picture.

According to Roy Chowdhury et al. (2025), the risks that arise at inference time differ from the risks encountered at training time in a fundamental way, which most work on privacy has concentrated on. Previous risk mitigation approaches to address what happens to sensitive data included in the training data of a model (e.g., differential privacy) or through federated learning. However, they do address the risk of consequences when a user enters their salary, medical diagnosis, or student number directly into a prompt. Preempt addresses this by classifying sensitive tokens and providing the relevant assurance at the moment of submitting a prompt prior to it reaching the model. The technical distinction between inference-time and training-time risks made by Roy Chowdhury et al. is pertinent to this research; it helps clarify the inadequacy of using existing training-time protections alone and also supports the need for a prompt-level intervention concerning protecting sensitive information.

❖ Technical Approaches to Prompt-Level Privacy Protection

Casper is a promising solution for ensuring privacy when using AI models. Developed by Chong et al. in 2024, Casper operates as a browser extension and can be run locally, without requiring the deployment of any changes to the underlying AI service itself. The design of Casper can be described as the most complete prompt sanitization system to date. The three-tiered sanitization mechanism of Casper consists of three separately defined component behaviours:

- Structured PII filtration by rule-based framework (i.e., email address, phone number, etc.)
- Contextual entity recognition using machine learning-based Named Entity Recognition (NER) model, and
- Use of locally executing browser-based large language model (LLM) to classify privacy-sensitive topics, which do not fall under the previous two PII-defined classifications.

The accuracy of Casper's PII detection was measured at 98.5% and 89.9% accurate for the identification of sensitive topics, based on a sample of 4,000 synthesized prompts evaluated. The challenge with Casper is that it is built and tested on prompts from general users, and, therefore, its topic classifier has not been trained to detect sensitive content that

exists within the context of academia. Furthermore, students often include information related to institutions when requesting assistance with coursework; thus, the prompts submitted by students to Casper are not structurally coded to cater to sensitive institutional information requirements.

Asthana et al.'s (2024) research vies to quantify the extent of the differences between context-aware and context-blind detection methodologies. The Adaptive Policy Engine, which translates compliance mandates into corresponding detection rules; the Dynamic Contextual PII Detection Module, which evaluates the connection of tokens to their immediate environments instead of assessing them solely on an individual basis; and the Adaptive Masking and Anonymization Techniques that correspond to each mandate are the three elements of an Adaptive PII Mitigation Framework.

According to Mancera et al. (2025), named entity recognition (NER)-based anonymization successfully protects user privacy in high-stakes AI applications while maintaining system performance. The PBa-LLM framework was tested using a corpus of 24,000 candidate profiles with two language models (BERT and RoBERTa) and six anonymization algorithms, and demonstrated that techniques that were privacy-preserving did not affect the performance of the underlying models. The principle transfers directly to the student prompt domain: it is technically feasible to identify sensitive attributes in naturally written text and neutralize them without destroying the usefulness of the system

❖ The Contextual Nature of Sensitive Data

Multiple sources in this paper point to a conclusion that will influence how any system that detects sensitive information should be created: sensitive data is not solely based on the data itself, but also on the context surrounding the data. The authors Telkamp and Hulsebos (2025) make this point the most recently. Their research has identified two methods for identifying the context in which sensitive information or data exist. The first approach is called type contextualization, which can identify the type of a data point based solely on content and then, once the database has determined what type of content it is, they will check to see if there is actual sensitive data (i.e. confidential information) around that content by looking at the complete context of the document. The second method is called domain contextualization, where the system looks to external sources/knowledge to determine what type of data may be sensitive due to its circumstances of use. When using type contextualization, the authors found the number of false positives was greatly reduced compared to commercial systems. Type contextualization had a recall of 94% versus 63% for commercial systems. Given the difference in recall value, there are 31% more instances of sensitive information than will have been found by keyword-based systems.

Telkamp et al. (2025) provide real world examples from data sets that may be used by organizations in an emergency response; specifically, uses for geographical locations of facilities, ethnical identifiers, and military related activity; that the sensitivity of those locations/facilities, or identifiers may depend on external geo-political conditions and also when the data are used. In the case of student prompts, an example would be when a student is describing information about his/her struggles or mental health issues with regard to completing an essay and asks a teacher for assistance, the information provided by the student is confidential even if there is no name associated with that information.

The article by Villardar (2024) presents a technological architecture for understanding contexts. Semantic Decomposition divides a prompt or question into structured, hierarchical components called "information schemata" that can be processed by systems. Selective Context Filtering identifies and removes extraneous context data, while allowing relevant context information to be retained for accurate classification or identification of actions to be taken. These methods allow an NLP system to comprehend both the presence of words in a student's question, as well as their meanings relative to the other words, and thus understand the academic activity being conducted.

Nakka et al. (2024) expand previous work on prompt-level evaluation by providing a way to identify whether a prompt contains personally identifiable information (PII). Even when a prompt does not contain any PII, an adversarial agent with access to in-domain data can recover PII contained within the language system's stored history of training. This is because using relevant contextual data to ground answers significantly increases the likelihood of recovering PII from drive responses (10+ times) and can for example allow 1 in 15 phone numbers to be extracted from a prompt with 2,308 prompts. Finally, McCoy et al. (2021) noted the language-models themselves exhibit risk of memorizing by demonstrating that RAVEN had the ability to reproduce historical passages of greater than 1,000 words.

❖ The Gap This Research Addresses

All fourteen sources confirm the facts of the issue and create one clear gap between these sources. The Ethnic literature identifies the potential for risk and recommends intervention. The Legal literature confirms POPIA's backing of the duty to the institution. The Empirical literature illustrates the manner with which the risk manifests itself in practice. The Technical literature provides methods for finding and suppressing sensitive materials. The science of detection also identifies that contextually based methodologies are significantly superior to rule-based methods.

None of the sources, however, meet the criteria of incorporating all of the above sources into a single use case regarding higher education and student AI prompt interaction. There are currently no existing systems developed to identify and suppress sensitive information in student AI prompts, used in higher education settings, which utilizes a contextually aware NLP methodology and understands the context of the academic and institutional conditions under which those prompts are produced. The student context has its unique set of categories of sensitive information, as well as regulatory requirements under POPIA, and the behaviors of the users within that context are still being formed. This is the aim of this research and the gap it seeks to fill.

7. Research Design and Methodology

❖ Research Approach

This research will adopt a qualitative approach. This choice is influenced by the type of research problem, and the research questions as well. Measurement of the frequency of something occurring, is not the main concern; rather, the goal of this research is to understand the reasons of occurrence, student experiences, and their meanings associated with such experiences; and all of these types of questions necessitate that the researchers engage with human experience. Qualitative research is the most appropriate means of accomplishing this type of inquiry.

Creswell (2014) describes qualitative research as appropriate for conducting a study when a researcher seeks to clarify the meaning given to a phenomenon by the individuals experiencing the phenomenon; when the research subject requires exploration of rather than confirmation of the research theory; and when a current theoretical framework does not adequately explain an observable phenomenon. All three criteria are available in this instance. AI privacy awareness by students, the rationale behind students' prompt behaviors, as well as the students' responses to mitigation strategies that include the use of warnings and the rephrasing of prompts, cannot be readily quantified by only using numerical evidence; they must be examined through dialogue, exploration and/or interpretation.

The qualitative approach to studying this research topic is consistent with the interpretivist research philosophy, which is based on the assumption that reality is socially developed/customized and that to properly comprehend human behavior, researchers must engage with the individual subjective experience of human beings (Bryman, 2016). In this

research, that means understanding not just what students share through AI prompts, but what they believe, what they know, and what they feel about the process.

❖ Research Design

The research design to be applied in the study is an exploratory qualitative design. An exploratory design will be used when there has not been a significant amount of research conducted on the area of inquiry in a specific context, and when the purpose of the study is to generate insight into and understanding of the phenomenon, rather than to test hypotheses (Saunders et al., 2019). The convergence of how students prompt artificial intelligence with respect to using natural language processing and privacy protection in higher education in South Africa is a perfect example of this type of research. The literature in Section 6 demonstrates the broad extent of the problem, however, the students' experience under this specific context has not been well-studied.

An exploratory design provides the ability for the researcher to move in the direction indicated by the data collected. If the student data reveals unanticipated areas of concern or patterns of experience that were not covered in the literature, an exploratory design accommodates for that. Additionally, if the students' experiences point to challenges to the assumptions being drawn from enterprise-oriented technical literature, an exploratory design allows for those challenges to be explored. These characteristics are benefits of the exploratory approach, and given there is very little information available about the target population of this study, an exploratory design is an appropriate choice for this study.

❖ Justification for Qualitative Approach

Although a quantitative study will provide the frequency of sensitive content found in students' academic AI prompts, it is unable to provide the reasoning as to why that content is included, student's understanding of risks associated with sensitive material, nor indicate how students will respond when an AI system intervenes. Consequently, a qualitative approach will be applied to supply answers to the research questions being postulated.

The qualitative findings will guide the development of the context-aware NLP system within this research, but do not act as a means to evaluate it. Through interviews with students, the qualitative phase of the research will determine the types of sensitive content displayed in student AI prompts, descriptive/disclosure nuances of language, and types of interventions postulated to be acceptable and comprehensible by the students. These qualitative parameters will serve to construct the targets for detection and mitigation of sensitive content within the proposed system. Therefore, there is no separation between the

quantitative quality and technical development of the proposed system. Instead, both will be developed and clarified as a function of their interdependence throughout the entire research design process.

The relationship between qualitative comprehension and technical development is not new. Chong, Wong et al. (2024) mention that the topic classification function of Casper was created based on understanding the types of sensitive information that users have disclosed in real life. In contrast, Mireshghallah et al. (2024) created their classification based on qualitative analysis of the actual conversations prior to utilizing it for guiding detection method design. This project will conduct similar research on a different population and a different setting where research has not previously been completed.

❖ Research Quality: Trustworthiness and Credibility

In qualitative research, the criteria for evaluating research quality differ from reliability and validity standards in quantitative studies. Lincoln and Guba (1985) propose four criteria for trustworthiness in qualitative research: credibility, transferability, dependability, and confirmability. This research addresses each of these.

Credibility will be addressed through member checking: participants will be given the opportunity to review summaries of their interview findings and confirm that the researcher has accurately captured their meaning. Transferability will be addressed through thick description: the research context, participant characteristics, and data collection process will be documented in sufficient detail that readers can assess the applicability of findings to their own contexts. Dependability will be addressed through a detailed audit trail of methodological decisions. Confirmability will be addressed by grounding all interpretations in direct quotations from the interview data, making the evidential basis of each finding transparent to the reader.

8. Population, Sample Method and Sample Size

❖ Study Population

The sample will consist of students enrolled in all faculties at the Cape Peninsula University of Technology (CPUT) who have utilized AI Tools such as ChatGPT, Google Gemini or Microsoft Copilot for academic purposes. CPUT was chosen as the site for this research because it is the institution the researcher is affiliated with, thus providing easy access to the sample, and because of the diverse background (across faculties, disciplines and socioeconomic status) of the student population, which will provide richness to the qualitative data that will be collected.

The focus on students currently utilizing AI tools is intentional as students who do not utilize such tools would not be impacted by the risks this research study addresses and would not provide useful information to answer the queries posed within this research study. Restricting the sample to CPUT is a practical boundary as well as a scope appropriate boundary. In addition, while the results of this study will be of most direct application to the CPUT context, the basic dynamics associated with student AI prompt behavior and privacy awareness will likely generalize similarly to institutions with similar characteristics.

❖ Sample Size

To conduct the semi-structured interviews, 20 to 30 undergraduate students at the selected college or university will be invited to participate in this research. The number of participants will be consistent with the guidelines for qualitative research using interview methods, where sample sizes are determined based on data sufficiency (i.e., interview data saturation occurs when the new data is largely redundant with that already collected, and no new themes/variations are emerging within the data) (Braun & Clarke, 2006). The approximate number of participants in the proposed focused exploratory study is sufficient to provide the breadth and depth of insight necessary to adequately address the research questions.

❖ Sampling Method

Participants will be recruited through convenience sampling. The participants in convenience sampling are selected on the basis of their willingness to participate and their easy accessibility (Saunders et al., 2019). In this particular study, the participants will be made up of CPUT students that have been made available to the researcher through the researcher's academic networks at CPUT, faculty noticeboards and student communication channels as well as through a volunteer sample.

There are a number of reasons as to why convenience sampling is an appropriate sampling method for this research study, the research is an exploratory, qualitative study, and therefore the focus of the research is depth of insight and understanding rather than statistical representation. The population of interest (CPUT students who utilize AI tools) do not have a formal record of their existence nor can they be identified as a discrete population; therefore, it is not feasible to use probabilistic sampling methods. The methodology of convenience sampling allows the researcher to recruit study subjects quickly and easily, but at the same time has procured the basic inclusion criteria of being a CPUT student actively using AI tools for academic purposes.

While there is a limitation to the findings of convenience sampling (i.e., the findings cannot be statistically generalized to the general population), such a limitation is acknowledged and is appropriate for the scope and focus of the study. Qualitative findings are not meant to be generalized in a statistical sense; they are meant to be transferred from one context to another, which means that the reader can assess the applicability of the findings for their own context (Lincoln & Guba, 1985).

❖ Inclusion and Exclusion Criteria

Inclusion of participants will be limited to students currently studying at CPUT and who meet the following requirements: they are enrolled at CPUT in any faculty, they have used at least one AI-based tool for academic purposes within the last 6 months; they are 18 years of age or older; and they are prepared to provide informed consent to participate. Students who have not used AI-based tools for academic purposes will be ineligible to participate in the study as they cannot provide information to address the research questions. In addition, students under 18 years of age will not be allowed to participate due to the additional ethical and consent requirements that must be adhered to by researchers under South African law when conducting research involving minors.

9. Data Collection Instruments and Fieldwork

❖ Research Instrument: Semi-Structured Interview

The data for this study will be gathered using semi-structured interviews conducted online via video conferencing tool. As such, using a semi-structured interview approach is an appropriate instrument for conducting the research, as it contains both the structure and consistency of a predetermined set of questions, as well as the flexibility for the researcher to pursue directions that arise from participant responses to questions being asked. In contrast to a structured questionnaire, researchers using a semi-structured interview will be able to ask follow-up questions as needed to probe deeper into a topic when a participant raises an unexpected topic. In contrast to unstructured conversations, the investigator will be able to cover all core topics necessary to answer the research questions with all participants.

Bryman (2016) has described the semi-structured interview as ideally suited for qualitative research where the researcher seeks to understand participants' meanings, attitudes, and experiences, which is the goal of this study. The thematic categories utilized in this study will include:

- Ways in which the participant has used AI previously and for what tasks AI has been used
- The participant's awareness and understanding of potential data privacy risks associated with using AI
- Whether or not the participant can remember any incident in which he/she provided personal/sensitive information via a prompt, and
- How the participant would respond in three potential scenarios when prompted by the AI tool (i.e., whether or not the AI 'warned' the participant, recommended a rewording of the prompt, or requested an indication of whether the participant acknowledged that there could be a data privacy or security risk).

❖ Interview Conduct and Mode

To make it easier for participants from the Cape Town area to take part in the interviews and to reduce any logistical hurdles associated with participating, all interview participants (i.e., P) will complete their interviews online via video call by using their own computer. Video call interviews are widely accepted as a proper and legitimate technique for conducting qualitative research and can provide data of comparable quality to data obtained from face-to-face interviews, as long as researchers maintain an active level of interest and engagement with their interview participants during the interview process and interview participants sit in a comfortable and private location during their interviews (Braun & Clarke, 2006).

Interviews are expected to take 30-45 minutes to complete based on the Participant's expectation of the interview. With the agreement of the Participant, the video call platform will allow for the recording of the audio portion of the interview and the verbatim transcribing of the recorded audio for the purpose of subsequent analysis. Participants who do not agree to have their interview audio recorded will not be excluded from participating in the study. Thus, for participants who do not agree to have their interviews audio recorded, comprehensive notes will be documented throughout the time of their interview and immediately following the completion of the interview to capture the most accurate representation of core themes identified or quotes cited by each interview Participant at the time of their interview.

❖ Piloting and Instrument Refinement

Before the start of formal data collection, the interview guide will be piloted with three to five CPUT students who will not be included as part of the main sample. The goals of piloting are to evaluate how easily respondents can understand and respond to each question, to

identify any questions that might generate confusion or misinterpretations, and to provide an estimate of how long an average interview will take to complete. The results from the pilot study will be used to improve upon the interview guide prior to commencing with the main data collection.

10. Data Coding and Analysis

❖ Thematic Analysis

Thematic analysis will be utilized to analyze qualitative data gathered through interviews, using the six phases as set forth by Braun and Clarke (2006) (i.e., **phase 1**: Familiarizing yourself with the data, **phase 2**: Generating initial codes to apply to the data, **phase 3**: Searching for themes from the generated codes; phase 4: Reviewing themes against the interview texts; **phase 5**: Developing a formal definition for each theme and giving each theme a name and creating the final analysis). Thematic analysis is appropriate for this study because of its flexibility in accommodating the broad range of topics covered within the interviews themselves; it has no dependence on a specific theory; and it generates results that are both usable and accessible as opposed to being very abstract in nature. Additionally, Braun and Clarke's (2006) six phases will be applied consistently to the entire sample of interviews collected.

❖ Coding Process

Coding will start with the careful and multiple readings of each transcript to help establish a level of confidence with each of the transcripts prior to assigning any codes. Initial coding will be conducted using an inductive approach; therefore, the codes will emerge from the data rather than be imposed on the data, since they are based on existing literature. By using this approach, the researcher will ensure that the findings reflect the actual words used by the participants, rather than the researcher's expectations.

During the coding process, codes will be grouped together to form larger candidate thematic categories that span across multiple transcripts. After creating the thematic categories, they will be checked against the complete dataset to ascertain that they are logical, distinct from other categories, and have enough data to support each theme individually. Themes that cannot support themselves after this review will be either combined, separated, or removed. Once the final themes have been identified, they will be given a descriptive label and an explanation, along with example quotations from each of the interview transcripts that correspond to each theme.

❖ Connecting Analysis to Research Questions

The thematic analysis will be organized around the four specific research questions posed in the study. The first research question, which addresses student awareness of AI privacy risks, will be answered with emerging themes from participant accounts about their knowledge of AI Privacy Risk and its source(s). In regard to research question #2, regarding student types of sensitive information shared through AI prompts, themes will emerge from participant descriptions of their prompting and of prompting examples provided by the participant. Research question #3, regarding effective mitigation strategies will be addressed through emerging themes from participant reactions to the hypothetical interventions included in their interview guides. Lastly, the fourth research question (i.e., defined within a design methodology), regarding the design of a NLP model, will be influenced by the previously discussed themes and behaviors as interpreted through patterns observed regarding both student awareness and behavior (such as to develop detection targets or the logic for the proposed system).

11. Rationale, Significance and Demarcation

❖ Rationale for the Research

The motivation for this study arises out of three distinct but related realities. The first is how many students use AI technology and how quickly the number of AI technologies being used by both colleges and universities continues to rise, where the danger of exposing students' private data has been proven (Miresghallah et al. 2024) through an analysis of over one million real-world conversations. Second, as stated by Davis and Trott (2024), there exists a duty to protect a student's private data; therefore, the requirement for institutions to protect students' data is clearly documented through the analysis of South Africa's Protection of Personal Information Act's (POPIA) application to AI systems. Third, there does not exist a technical system that has been developed or designed to address these unique circumstances, a realization supported by the entire literature cited in this study. Thus, this Research Study is warranted because there is an actual issue impacting students, academic institutions have a legal duty to protect student data and a technical solution does not currently exist.

In addition to the technical issues, there is a need for institutions of higher education to educate students regarding the use of AI and how this will ultimately impact their professional careers after graduating from college. Both Huang (2023) and Mishara (2024) note that students are developing habits around AI use while attending college and such developed habits will shape how they use AI in their respective careers.

❖ Significance of the Research

The value of this study exists at three different levels. From the perspective of the student, it offers a real-life protection mechanism for an otherwise exposed, unaware population. As it relates to institutions specifically, the study offers South African universities a tangible mechanism for fulfilling their obligations under POPIA as they pertain to the use of AI by students, thereby closing the regulatory gap that has been identified by Davis and Trott (2024). Additionally, the research creates a student-oriented implementation of context-aware NLP detection techniques, contributing to the foundational contexts provided by Villardar (2024), and Telkamp and Hulsebos (2025) in a context that has not yet been demonstrated or applied.

The research will also be of value to AI developers and policy-makers who are concerned with the ethical deployment of AI tools in educational settings. The results of this research will provide information about the ways in which students utilize and interact with AI tools, what their understanding of privacy risks is, and what types of interventions they consider acceptable and effective. Prior to this study, there has been little research published that would support the development of design decisions and policies regarding the use of AI in education; therefore, this research will fill a gap in the current research literature.

12. Ethical Considerations

❖ Informed Consent

All the participants should have provided adequate written consent prior to beginning participation in a proposed study. "Adequate informed consent," or simply, "informed consent," indicates that participants have received an adequate amount of clear, accurate, and complete information regarding: 1) The purpose of the study; 2) What will be required of them through their involvement in the study; 3) Possible risks and benefits of their participation in this research study; 4) Their right to withdraw from the study at any time without incurring any form of penalty or consequence (Republic of South Africa, 2013). The informed consent and any other documentation used to obtain participant consent will be written using plain language, easy to read, and will be made available to prospective study participants prior to scheduling an interview.

Consent for participation in online interviews will also be obtained through written communication (e-mail) or through an electronic signature before each scheduled online interview. At the start of each online interview, verbal consent will also be documented in the interview notes. The decision to participate or not in the study is completely voluntary. No participant will feel pressured in any way, nor will receive incentives to participate beyond

a reasonable acknowledgement of their time, nor will they in any manner be penalized by their decision to participate.

❖ Anonymity and Confidentiality

None of the research outputs from this research project will reveal the identity of participants. Therefore, there will not be any identifiable information contained within any published or shared document which could identify their participants in any way. Before the analysis of their interviews (i.e., the transcript), pseudonyms will be assigned to all participants on a consistent basis through all research outputs and stored in a separate secure storage facility from the research data on the basis of confidentiality (Bryman, 2016). This storage facility will be protected by a password that only the principal research will have access to. At the time of data collection, all participants will be informed of this commitment and will be reminded at the beginning of each session that anything they share with this research will remain confidential.

❖ Right to Withdraw

All research participants have the option to remove themselves from any portion of the study, including after their interview has been completed, even if they do not wish to provide an explanation for their removal. If a subject removes themselves from the study following their interview, the researcher will eliminate any data collected about the subject from the research sample and will not utilize that data for the analysis. This will be made very clear on the consent form, and it will be stated again at the beginning of each interview.

❖ Data Storage and POPIA Compliance

All research-related data will be maintained on a CPUT owned electronic device and secured by a password according to their data governance and the requirements of POPIA and the Republic of South Africa's rights to process personal information (South Africa, 2013). The audio recording files created from participant interviews will be erased within 30 days after transcription; however, the pseudonymized transcript will be kept for five years from the completion date. After five years, pseudonymed transcriptions will also be securely destroyed. All participant-related data will never be stored on third-party cloud services or shared with anyone else except for research team members without prior written consent from an individual participant.

❖ Ethical Approval

Prior to commencing data collection, the researcher will submit an application for ethical review and approval to the Research Ethics Committee (REC) at the Cape Peninsula University of Technology (CPUT). This research involves no deception; there are no physical risks involved with participating in this research; and there is no interaction with vulnerable populations beyond what would normally occur when conducting research on the general

student population. The main ethical risks of this research involve the confidentiality and privacy of participants' data, and both these issues will be adequately addressed by the processes described above and approved by the REC at CPUT. All research will be conducted in compliance with CPUT's Research Ethics Policy.

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